

Set-Period vs. Multiset-Period: Computational Results

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Setup

For each row (o_n, o_d, a_n, a_d , period) of the file `tests/new_find_patterns_x_max_1000000_51012_lines.csv`, the program `src/c/set/add_period_set` computed the set-based period length λ_{set} on the same parameter quadruple and appended it as a sixth column `period_set`. The run was performed in *global* mode with $X_{\text{max}} = 10^6$ and a per-row wall-clock cap of 120 seconds. The output is written to `tests/new_find_patterns_x_max_1000000_51012_lines.csv.tmp`.

Aggregate counts

Of the 51,011 data rows in the input, the resulting file breaks down as follows.

Outcome	Rows	Meaning
$\lambda_{\text{set}} = \text{period}$	50,968	set period equals multiset period
$\lambda_{\text{set}} = -2$	29	ARRAY_SIZE_EXCEEDED
$\lambda_{\text{set}} = -3$	1	DX_LENGTH_TO_SMALL
$\lambda_{\text{set}} = -4$	13	per-row 120 s timeout
$\lambda_{\text{set}} > 0$, $\lambda_{\text{set}} \neq \text{period}$	1	genuine disagreement (see below)
Total	51,012	

In other words, on 99.92% of the rows (50,968/51,011) the set-based and multiset-based period lengths coincide; the remaining 43 rows split into 43 computational aborts (out of memory, dx-window too small, or timeout) and exactly one row in which the two periods are both well-defined positive integers but disagree.

Interpretation under Theorem 6.1. The high agreement rate is structural, not coincidental. The input file `new_find_patterns_x_max_1000000_51012_lines.csv` is concentrated in the regime $N < D$, in which Theorem 6.1 of the companion paper `rational_cut_and_project_gap_periods.t` forces $\lambda_{\text{set}} = \lambda_{\text{multiset}}$. The lone positive disagreement therefore must lie in the complementary regime $N \geq D$, where the theorem predicts $\lambda_{\text{set}} = 1$ — and indeed it does (see the next two sections).

The single positive disagreement

The only row with $\lambda_{\text{set}} > 0$ and $\lambda_{\text{set}} \neq \text{period}$ is

o_n	o_d	a_n	a_d	multiset period	λ_{set}
98,281	139	68	149	153,431	1

Here the multiset period is 153,431 while the set-based period equals 1. The next section shows that $\lambda_{\text{set}} = 1$ is the *correct mathematical value* for this row, predicted by Theorem 6.1 of the companion paper `rational_cut_and_project_gap_periods.tex`: the parameters lie in the regime $N \geq D$, where the theorem forces $\lambda_{\text{set}} = 1$.

Why $\lambda_{\text{set}} = 1$ is correct for this row

The reported value is the mathematical answer predicted by Theorem 6.1 of `rational_cut_and_project_gap_periods.tex` and the heuristic in `src/c/set/mathematics.c` arrives at it correctly. Both perspectives are reviewed below.

Theorem 6.1 prediction. For this row $\alpha = 68$, $\beta = 149$, and $\omega = \gamma/\delta = 98,281/139 \approx 707.06$. Hence

$$D = \alpha^2 + \beta^2 = 4,624 + 22,201 = 26,825,$$

$$N = \lfloor \omega\alpha \rfloor + \lfloor \omega\beta \rfloor + 1 = 48,079 + 105,351 + 1 = 153,431,$$

so $N/D \approx 5.72 > 1$, i.e. $N \geq D$. Theorem 6.1 then yields $\lambda_{\text{set}} = 1$, exactly the value reported.

Why every integer is a projected value. The condition $N \geq D$ is equivalent to: every residue class modulo D is hit by at least one accepted lattice point. Hence the underlying set of projected values $\beta x + \alpha y$ is all of $\mathbb{Z}$, and the gap sequence between consecutive distinct values is identically 1.

Parameters and window. With $\gcd(\alpha, \beta) = 1$ and $\gcd(\gamma, \delta) = 1$ as above, the width of the cut window in y is

$$\frac{\gamma(\alpha + \beta)}{\beta\delta} = \frac{98,281 \cdot 217}{149 \cdot 139} = \frac{21,326,977}{20,711} \approx 1029.74,$$

so each x contributes roughly 1030 valid integer y 's — a very wide strip. The actual run picked $x_{\text{max}} = 1000$, producing 1,029,738 raw projected values $\beta x + \alpha y$. This wide-strip regime is exactly what makes $N \geq D$ here.

Sorted-difference structure. After sorting and dropping zero gaps (the set-collapse step in `lambda` when `sort = true`), the sequence of distinct sorted differences has 237,813 entries:

- 232,589 entries are equal to 1 ($\approx 97.8\%$),
- only 5,224 entries are different from 1 (and they take values in the small set $\{2, 3, 4, 7, 10, 13, 16, 29, 42, 55, 68\}$),
- the non-1 entries are concentrated entirely in the first ~ 6000 and the last ~ 6000 positions; the middle 95% of the array — positions 5971 through 231,821 — is one uninterrupted run of 1's.

Because $\gcd(\alpha, \beta) = 1$ and the cut strip is wide, the projected lattice $\beta x + \alpha y$ *saturates* the integer line in the bulk: consecutive distinct projected values differ by 1 across the entire interior. Only the two ends of the sorted dx-range — where the strip “runs out” — produce gaps larger than 1.

The trim-and-spiral heuristic locks onto the bulk. The period-detection logic in `src/c/set/mathematics.c` starts by trimming $(1 - \text{FRACTION_OF_REMAINING_ELEMENTS}) / 40$ of the sequence from each end (here 594 positions per side) and then spirals both ends inward by 1 each iteration, calling `find_period_length` on the surviving window, until either a period is detected or the window has shrunk by more than $1 - \text{FRACTION_OF_REMAINING_ELEMENTS} = 10\%$ of its initial size. `find_period_length` returns 1 exactly when every element of the window equals every other element.

For this row the spiral reaches the all-1 bulk after $k \approx 5397$ inward steps: at $k = 5397$ the window is `[5991, 231, 821]`, which lies entirely inside the bulk run `[5971, 231, 821]` and is therefore constant equal to 1. The algorithm reports $\lambda_{\text{set}} = 1$ and stops.

Interpretation. The bulk of the sorted-and-collapsed difference sequence is the constant 1 because the projected lattice $\beta x + \alpha y$ saturates $\mathbb{Z}$: for every integer $\tilde{p}$ in a long central range, at least one accepted lattice point projects to it. The boundary patches at indices `[0, 5970]` and `[231, 822, 237, 812]` are finite-window artefacts of computing on a bounded x -range; in the bi-infinite sequence they vanish and the sequence is the constant 1 everywhere. This saturation is exactly the geometric content of the case $N \geq D$ in Theorem 6.1, so the heuristic’s report $\lambda_{\text{set}} = 1$ is mathematically correct, not an artefact.

Run summary

- Resume from a partial run: 40,201 rows already present in the output were skipped; 10,811 rows were freshly processed.
- No hard failures during the resume run; 11 of the 13 timeouts shown above were produced in this run, the remaining 2 were already present in the partial output.
- Errors of types `-2` and `-3` accumulated 30 rows in total; these are computational limitations of the current `lambda` implementation at $X_{\text{max}} = 10^6$ and do not indicate a true mismatch of the two period notions.

Caveats

- Negative `period_set` values mean the set-based period was *not* computed, not that it differs from the multiset period. Excluding the 43 aborted rows, the multiset period and the set-based period agree on every single one of the remaining 50,968 rows except the row exhibited above.
- The timeout cap of 120s is heuristic; lifting it (or doubling `MAX_PERIOD_ARRAY_SIZE`) might convert some of the 43 aborts into successful computations.

Re-check of the 43 aborted rows

The 43 rows that produced negative `period_set` values in the first pass were extracted into `tests/errors_set_negative_lambda.csv` and re-run with three limits raised: `MAX_PERIOD_ARRAY_SIZE` from 10^8 to $4 \cdot 10^9$ (allocating ≈ 32 GB for the `dx` buffer), `X_MAX` from 10^6 to 10^9 , and the per-row wall-clock cap to 180 s. Output: `tests/errors_set_negative_lambda_recheck.csv`.

Outcome	Rows	Meaning
$\lambda_{\text{set}} = \text{period}$	26	newly resolved, set period equals multiset period
$\lambda_{\text{set}} = -4$	17	still timed out at 180 s per retry slice
$\lambda_{\text{set}} > 0, \lambda_{\text{set}} \neq \text{period}$	0	no new disagreement
Total	43	

Crucially, every one of the 26 rows newly resolved by the larger buffer agrees with the multiset period: the set-based and multiset-based period coincide on all of them. The remaining 17 rows are not disagreements; they are computations that the current implementation cannot complete in a 180-second budget per retry, because the retry loop in `add_period_set.c` doubles `target_points` each time it receives `DX_LENGTH_TO_SMALL` or `NO_PERIOD` and starts a fresh alarm window.

Theorem 6.1 resolution of the 17 timeout rows. Direct computation of $N = \lfloor \omega\alpha \rfloor + \lfloor \omega\beta \rfloor + 1$ and $D = \alpha^2 + \beta^2$ on the parameter quadruples of the 17 remaining timeout rows shows that *every one of them satisfies* $N < D$, with N/D ranging from about 10^{-4} to 0.4. Theorem 6.1 then forces $\lambda_{\text{set}} = N = \lambda_{\text{multiset}}$ for all 17 rows, where the multiset period is already known from the input file. No actual set-period computation is needed: the answer is mathematically determined by the multiset period, and the 17 timeouts represent redundant work that the C heuristic could skip entirely on input rows with $N < D$. Redesigning `add_period_set.c` to short-circuit on $N < D$ (read the multiset period directly) is a cheap implementation improvement but is out of scope for this report.

Updated picture for the full 51,011-row dataset

Combining the original pass and the re-check:

Outcome	Rows
$\lambda_{\text{set}} = \text{period}$ (confirmed agreement)	50,994
$\lambda_{\text{set}} > 0, \lambda_{\text{set}} \neq \text{period}$ (positive disagreement)	1
$\lambda_{\text{set}} = -4$ (still inconclusive at 180 s/retry)	17
$\lambda_{\text{set}} \in \{-1, -2, -3\}$ (still inconclusive otherwise)	0
Total	51,012

This raises the confirmed-agreement rate to $50,994/51,011 \approx 99.97\%$, with the lone positive disagreement $(o_n, o_d, a_n, a_d) = (98,281, 139, 68, 149)$, $\text{period} = 153,431$, $\lambda_{\text{set}} = 1$ unchanged from before. None of the 26 newly resolved rows produced an additional positive disagreement.

Under Theorem 6.1 the picture is structurally complete. Direct computation of N and D on every row shows that exactly one row of the 51,012 inputs lies in the regime $N \geq D$ (the lone disagreement above), and all 51,011 remaining rows satisfy $N < D$. The theorem then dictates: the 50,994 confirmed agreements are forced by $N < D$ (set and multiset coincide), the lone disagreement gives $\lambda_{\text{set}} = 1$ as predicted by the $N \geq D$ branch, and the 17 remaining timeouts also lie in $N < D$, so their set-period coincides with the multiset period already in the input file (see the post-mortem below). No further input row can produce a fresh positive disagreement that contradicts the theorem.